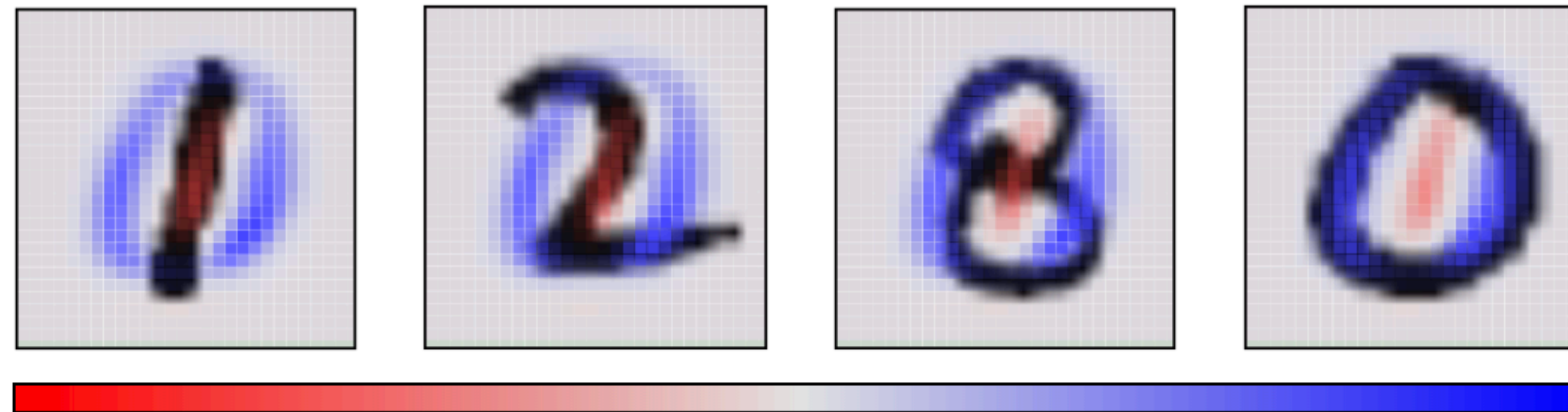
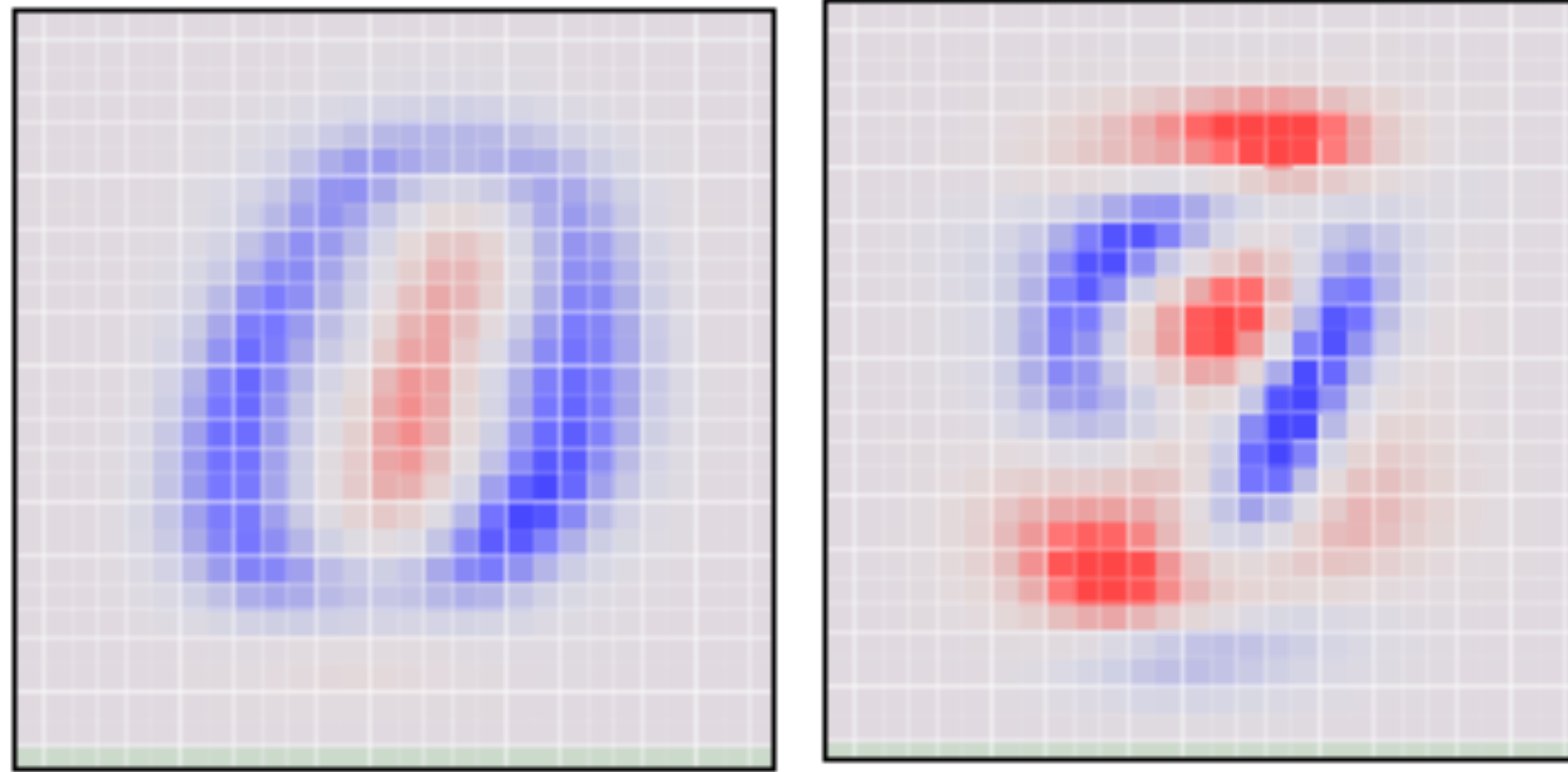


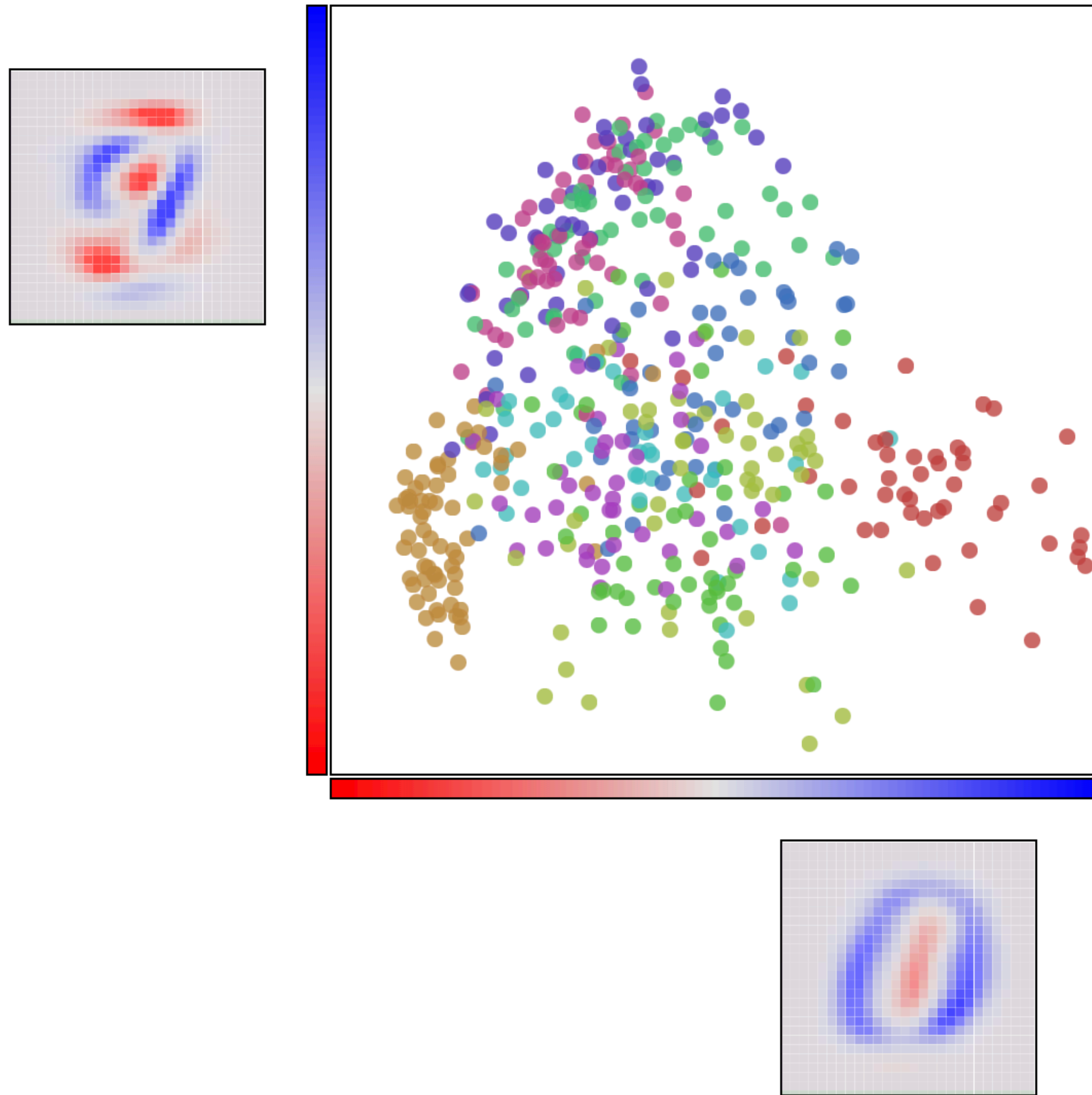
3장 : Homework + Chessbot

1. DATA 이해하기(내가 무엇을 시키는지)
2. 어떻게 학습시키지?(설계)
3. 아니 왜 안됨? (back22)

데이터 이해하기 - PCA

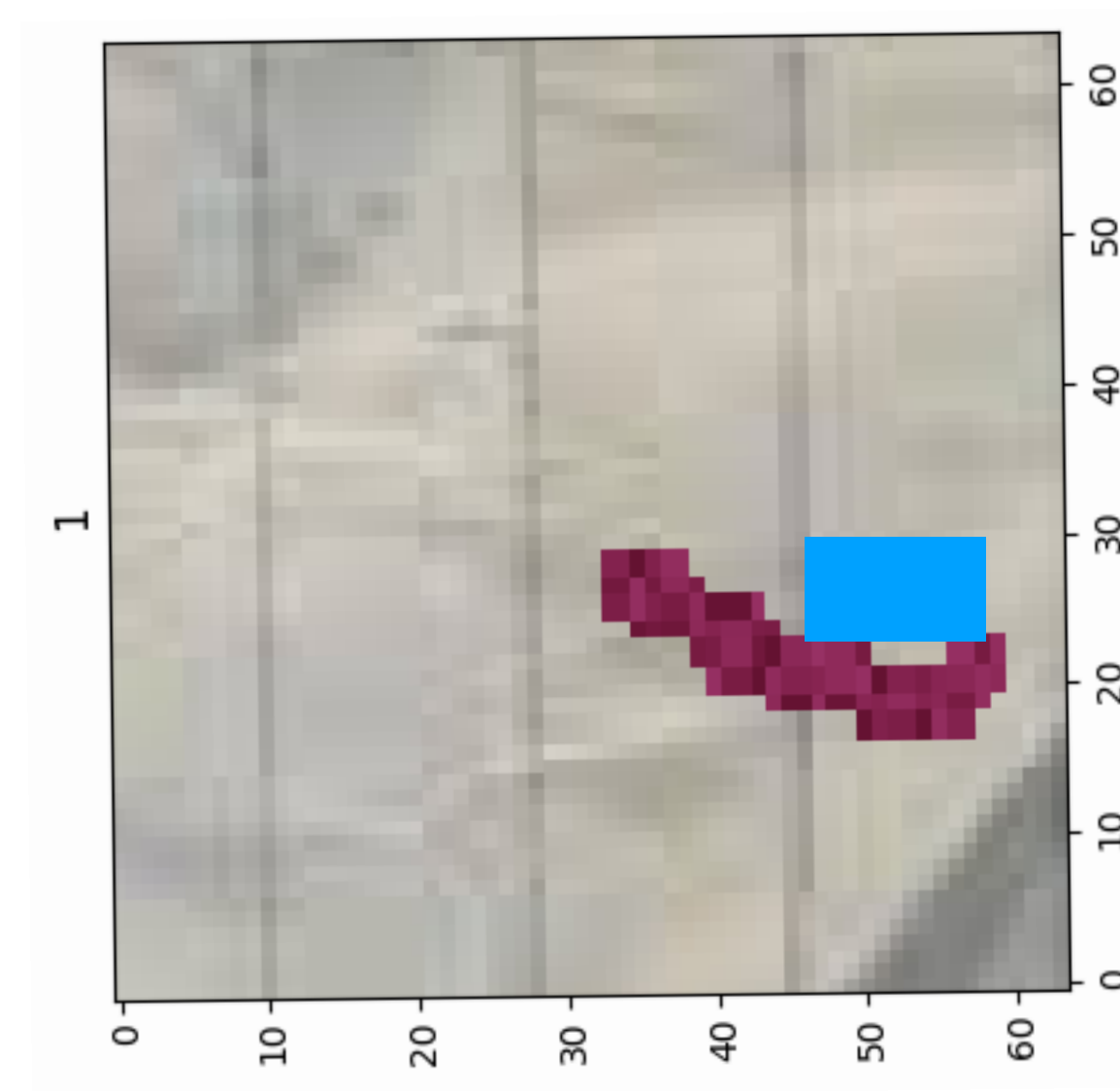
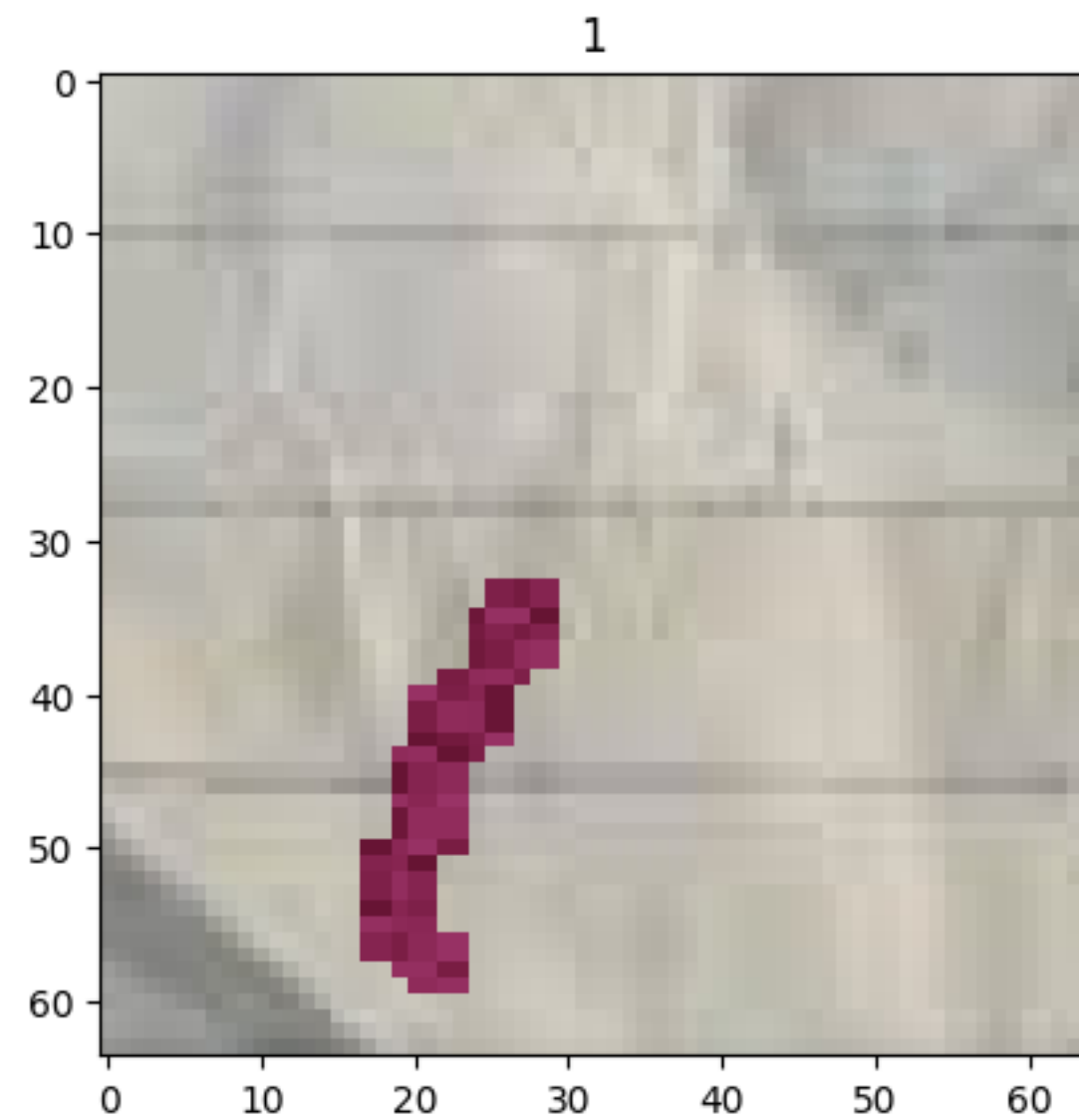


PCA



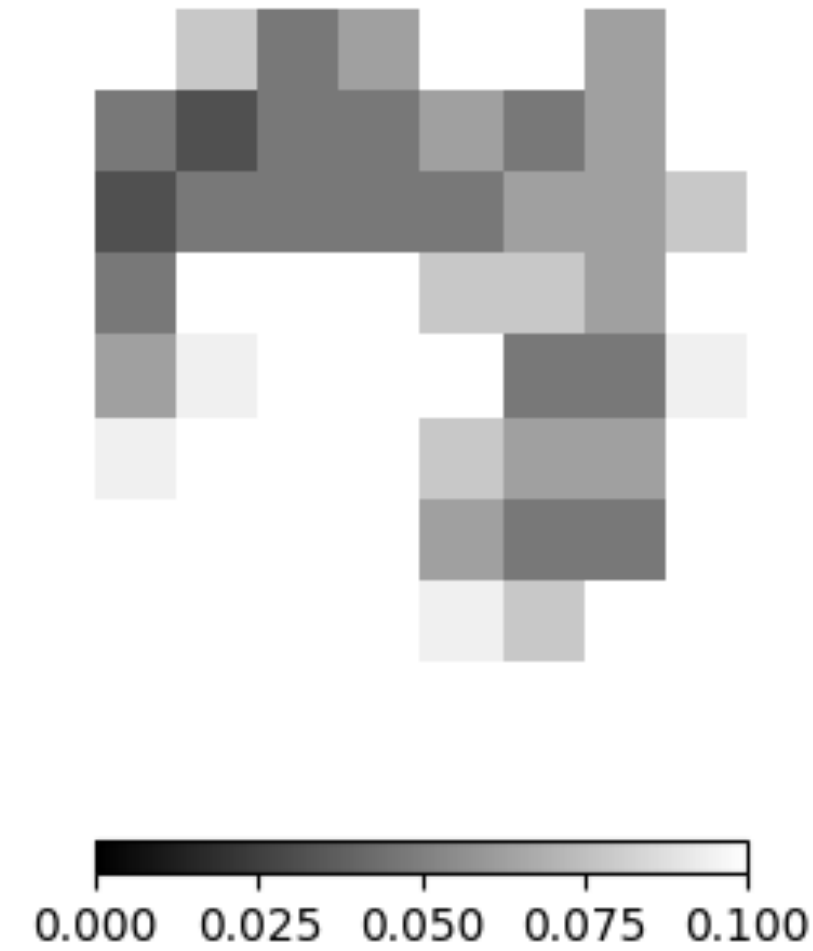
Visualizing MNIST with PCA

데이터 이해하기



뭘 배우니?

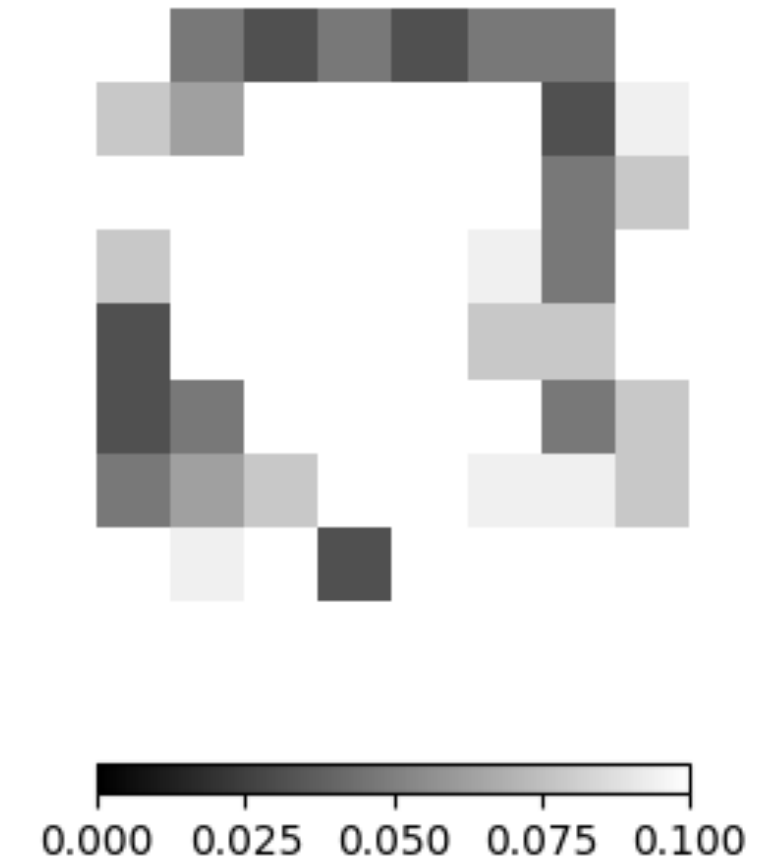
average act ratio of 64 channels(8x8)



1



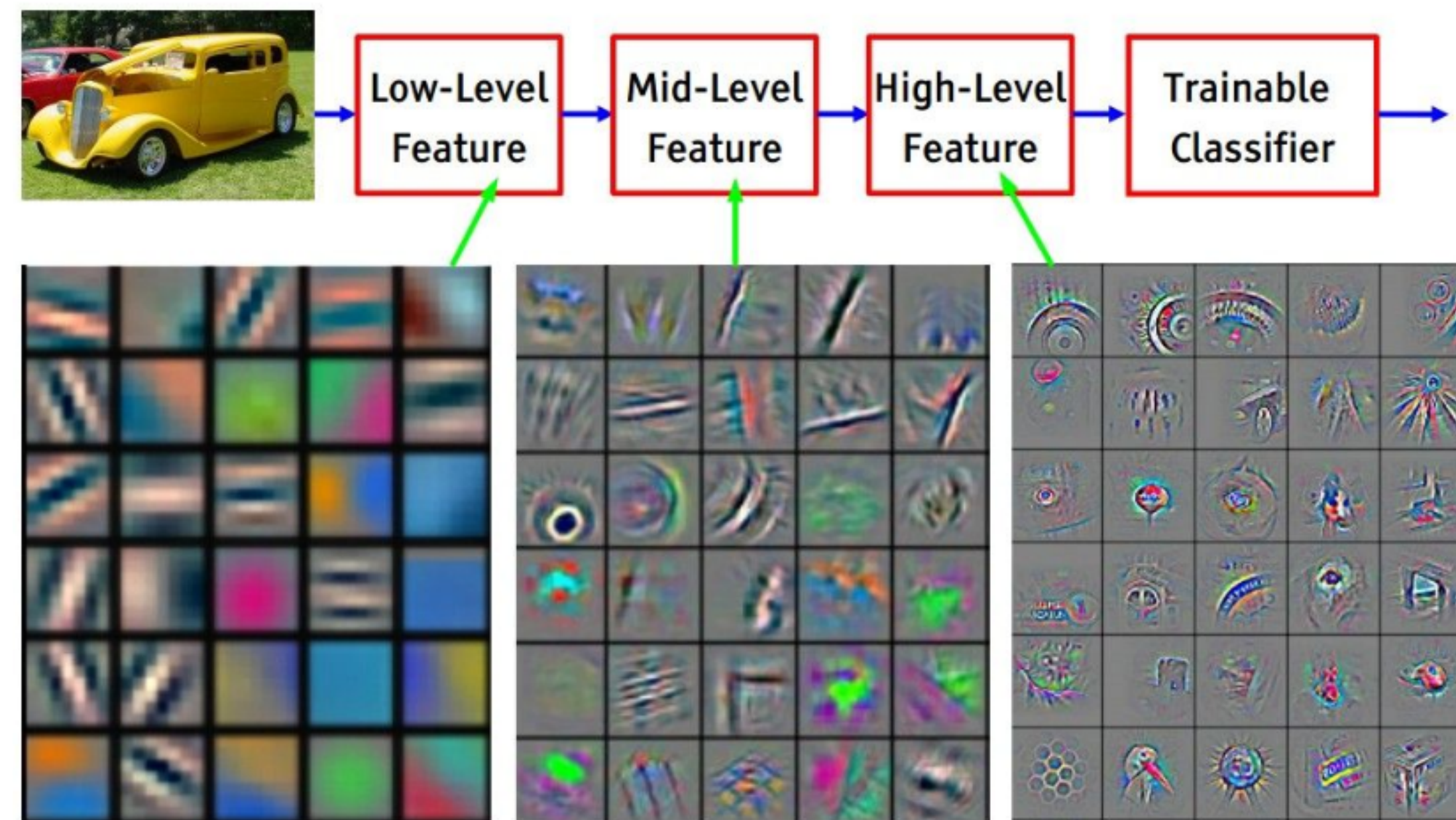
average act ratio of 64 channels(8x8)



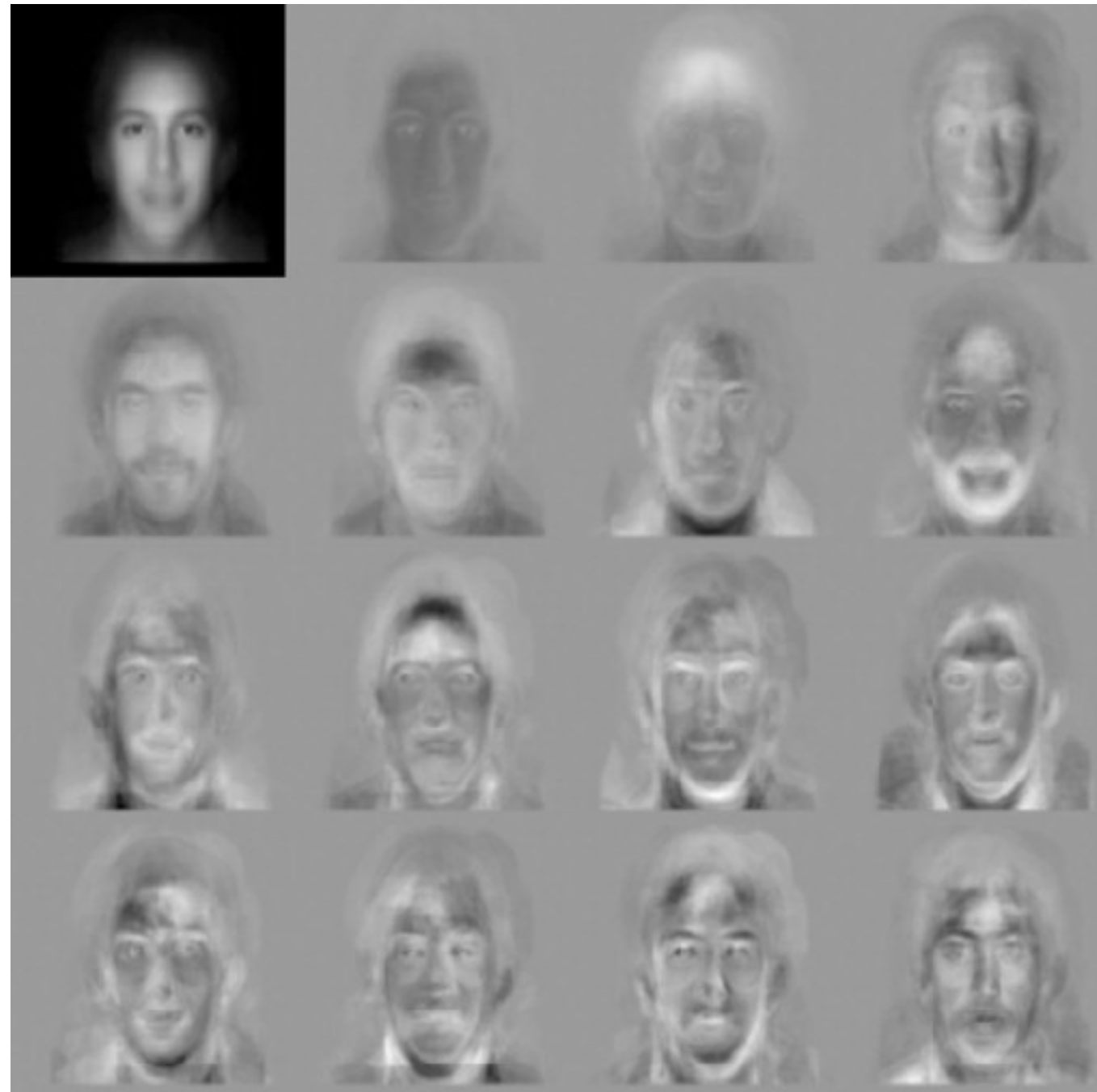
4



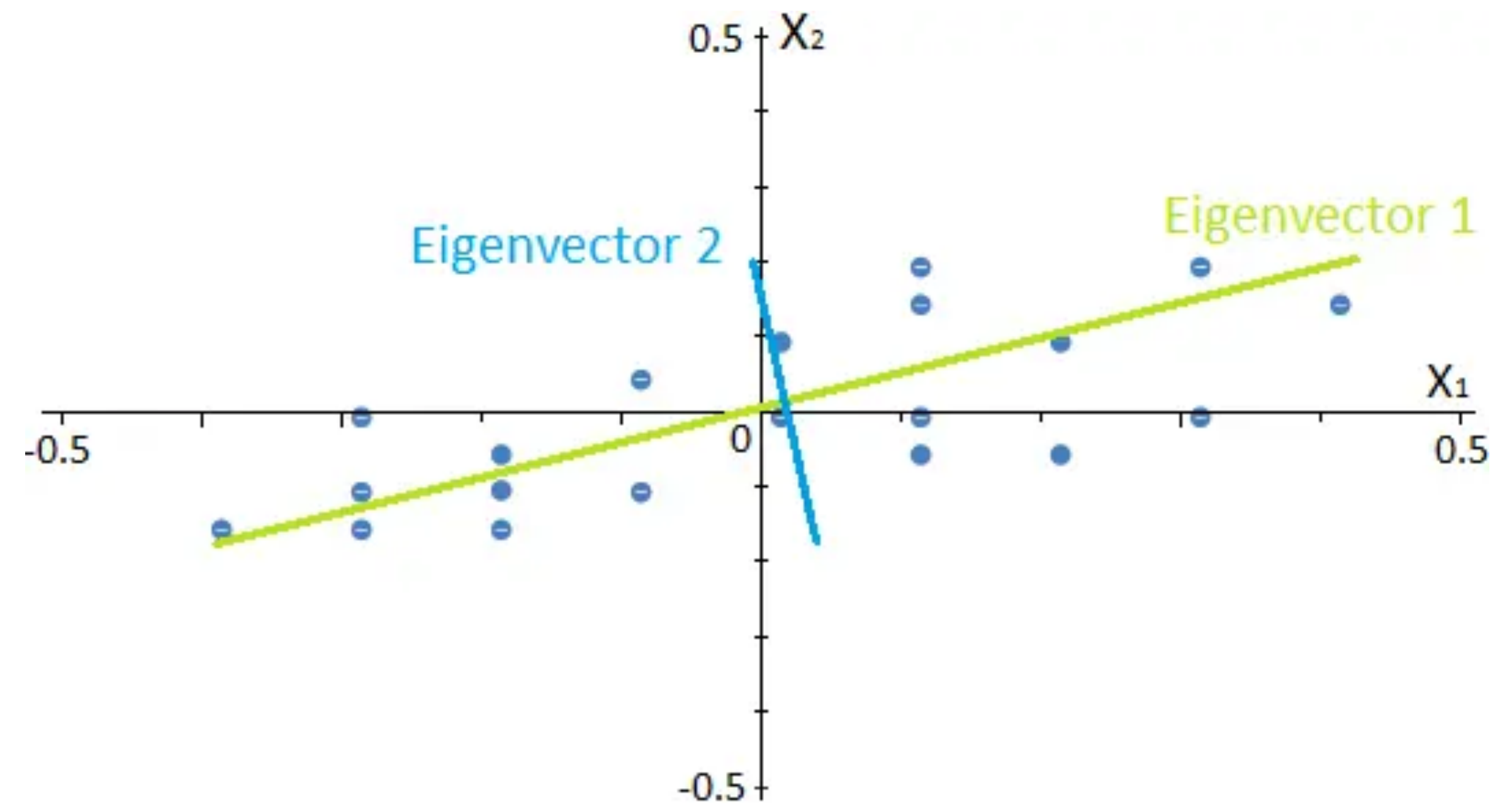
feature!!!요



feature!!!이요

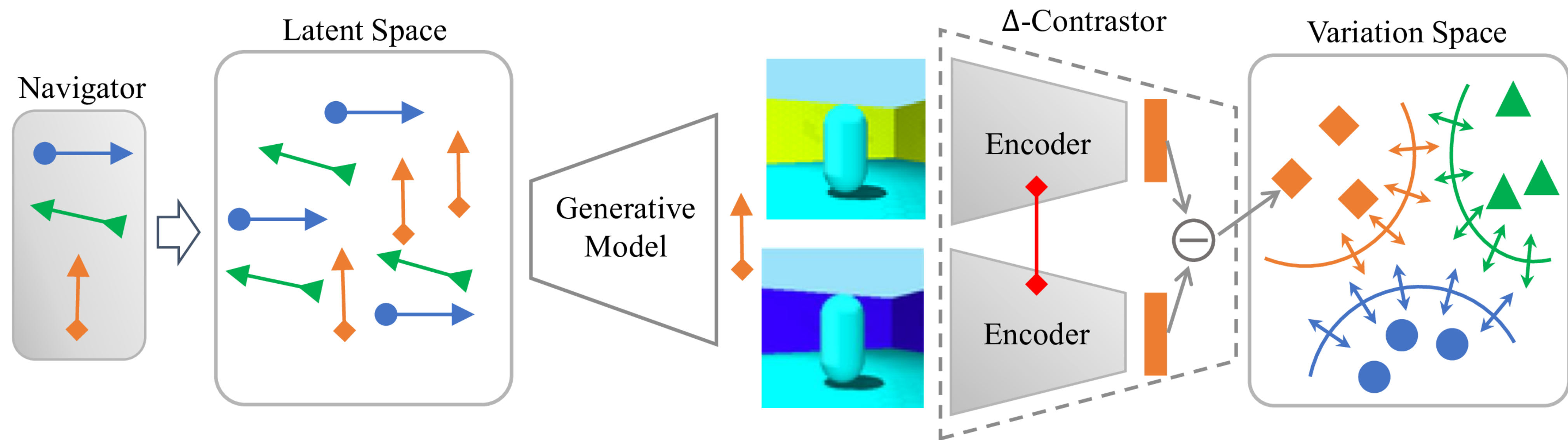


eigenface



capture high variance -> globally!!

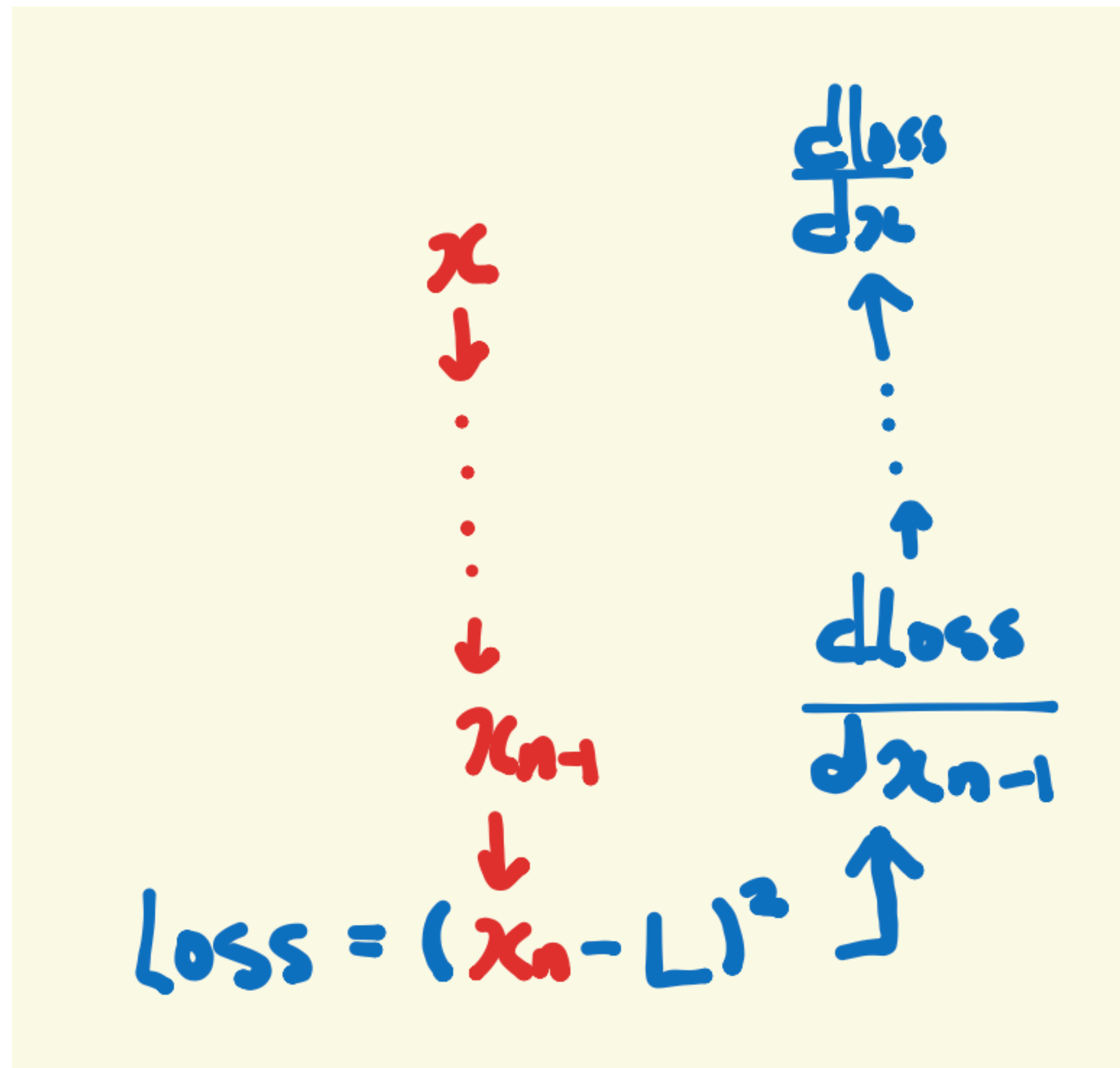
데이터 이해하기 - Disentanglement!



어떻게 배우게 하지?

- loss signal -> 교과서
- architecture -> inductive bias+stability
- optimizer -> 거의 Adam으로..

re: 인공지능은 어떻게 배우냐?



우리가 학습하고자 하는 m 과 x 가
얼마나 Loss를 화나게(미분) 하는
지를 알 수 있다.

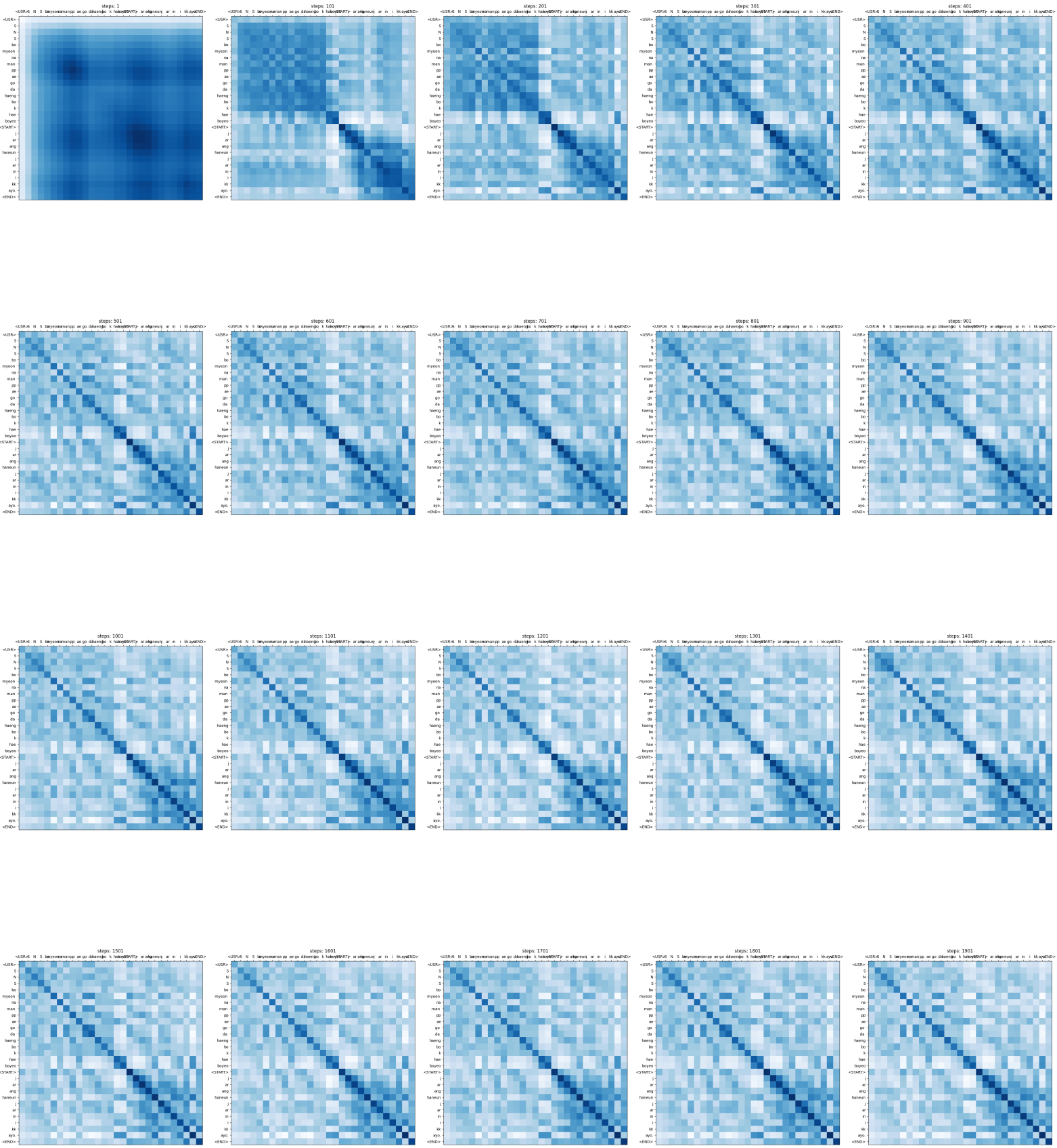
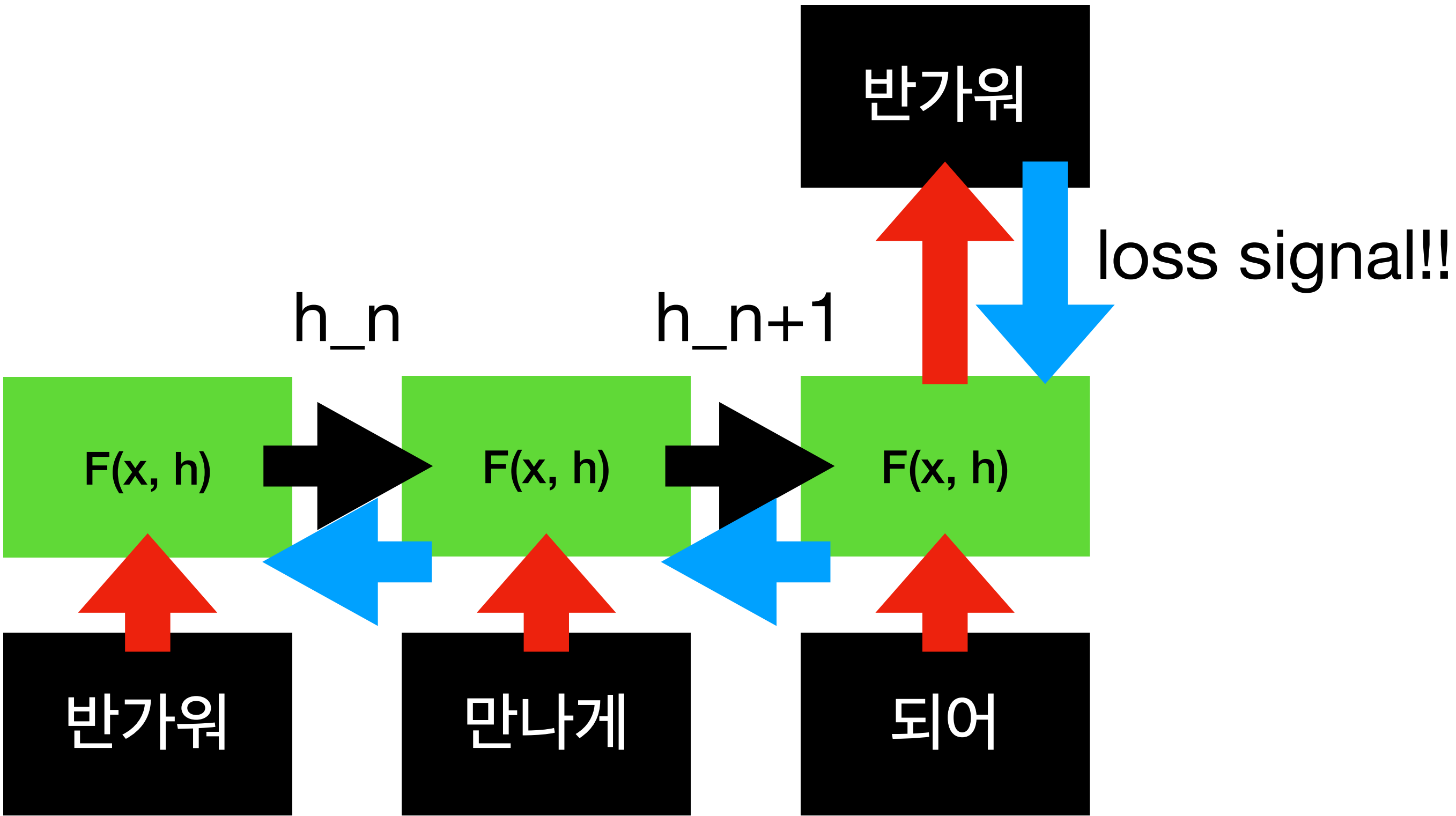


화나게 하는 방향과 반대로
+
화나게 하는 정도에 비례해서
값을 업데이트하자!

Architecture

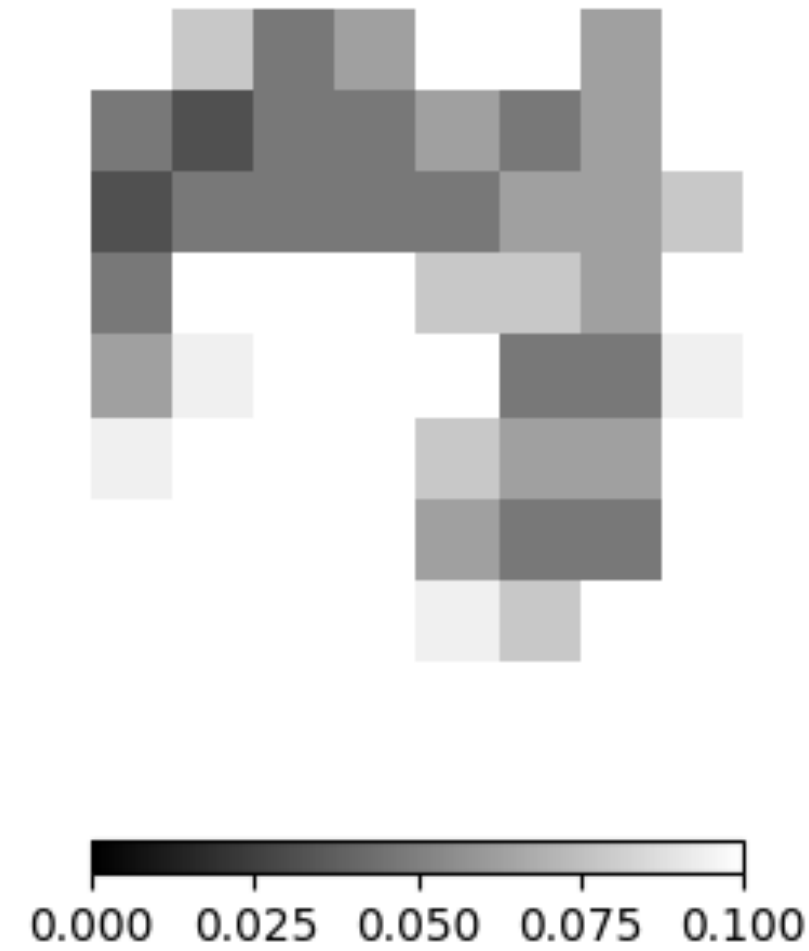
- ~~-loss signal -> 교과서~~
- architecture -> inductive bias+stability
- ~~-optimizer -> 거의 Adam으로..~~

Architecture



뭘 배우니?

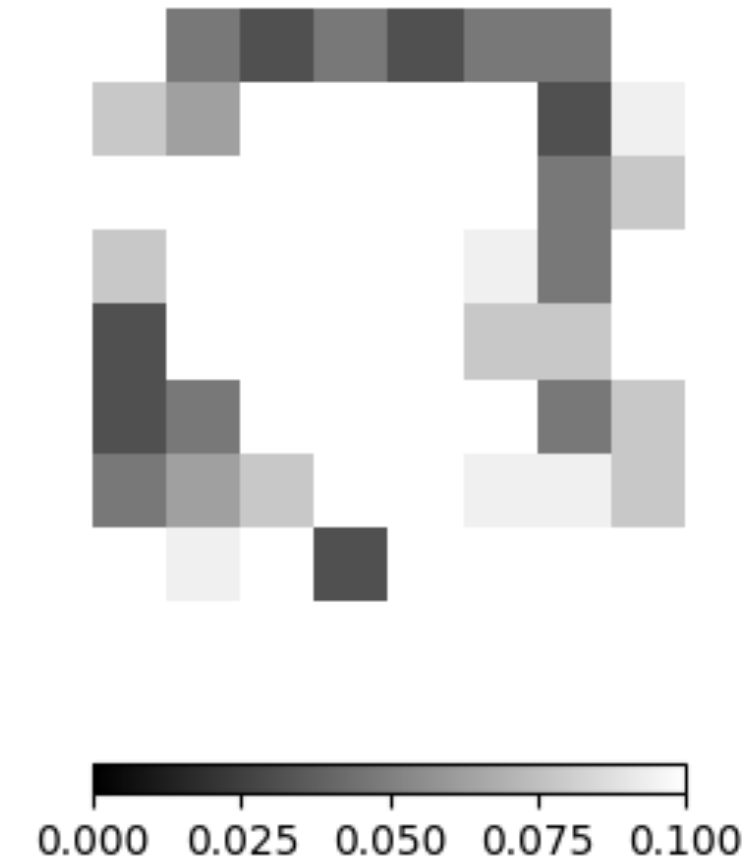
average act ratio of 64 channels(8x8)



1



average act ratio of 64 channels(8x8)



4



힌트를 줘봅시다.

Where are you giving attention?



체스 게임 데이터베이스

과거와 현재의 최강의 체스선수들의 고급 게임 수백만 가지를 검색하여 보세요. 세계 챔피언부터 FIDE 마스터에 이르기까지, 방대한 분량의 게임들을 검색, 분류, 그리고 다운로드 할 수 있습니다.



Garry Kasparov

2,420 게임



Bobby Fischer

1,194 게임



Magnus Carlsen

5,786 게임



Jose Raul Capablanca

1,310 게임

경기 기록 (1,017)

최근

일일

실시간

봇



전체: 1,017

< 1 2 3 4 ... > 마지막

플레이어	결과	정확성	수	날짜	
	0 1	리뷰	21	2025. 4. 22.	
	1 0	리뷰	31	2025. 4. 11.	
	1 0	리뷰	38	2025. 4. 10.	
	0 1	리뷰	28	2025. 4. 8.	
	1 0	리뷰	23	2025. 4. 7.	
	0 1	리뷰	12	2025. 4. 4.	
	1 0	리뷰	15	2025. 4. 4.	
	1 0	리뷰	24	2025. 4. 4.	

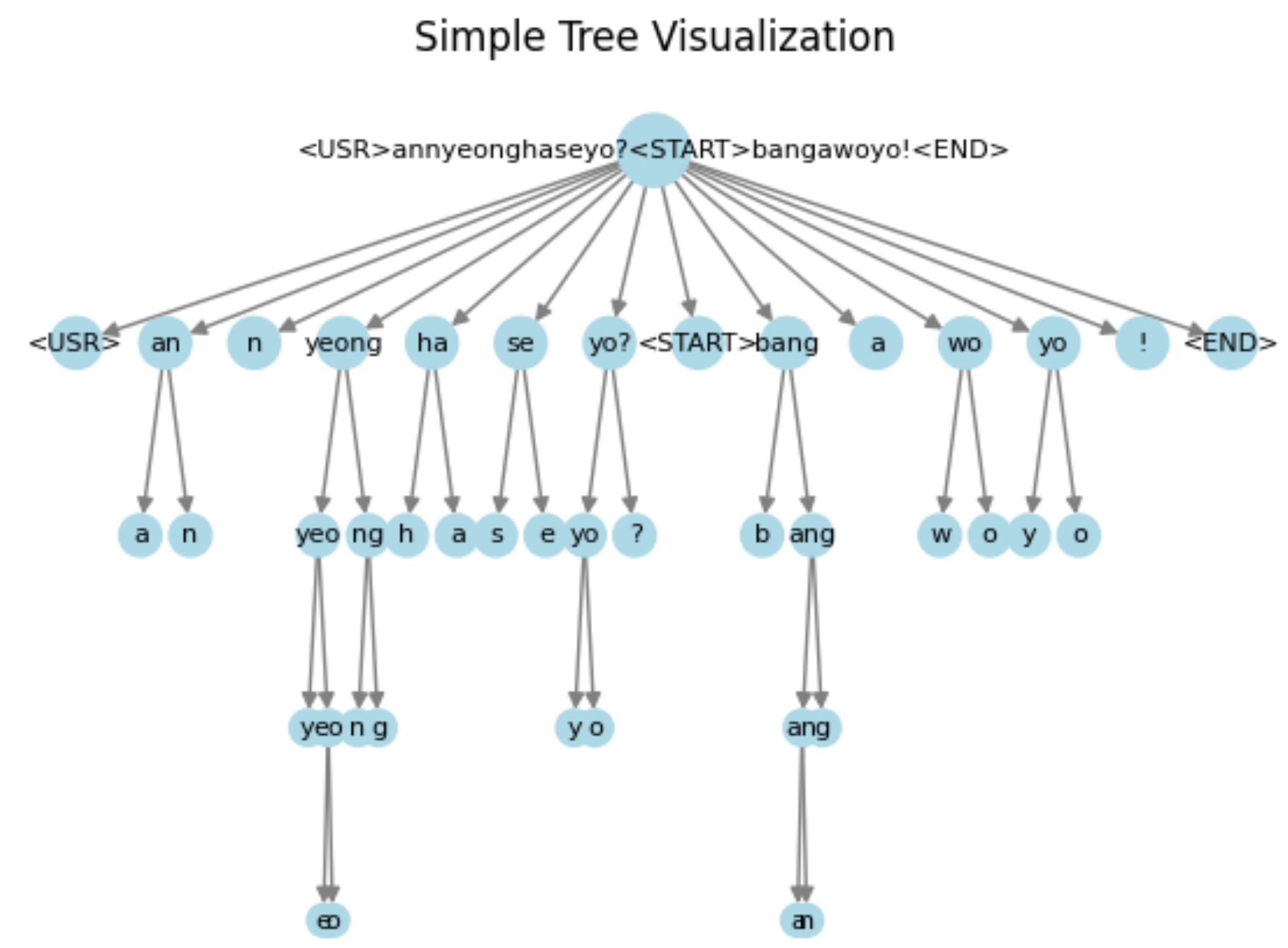
데이터 이해하기 - PGN

1. e4 {[%clk 0:01:00.9]} 1... e5 {[%clk 0:01:00.1]} 2. Nf3 {[%clk 0:01:01.5]} 2... Nc6 {[%clk 0:01:00.2]} 3. c4 {[%clk 0:01:01.3]} 3... Bc5 {[%clk 0:01:00.6]} 4. Nc3 {[%clk 0:01:01.5]} 4... Nf6 {[%clk 0:01:00.8]} 5. d3 {[%clk 0:01:00.7]} 5... O-O {[%clk 0:01:01.2]} 6. g3 {[%clk 0:00:57.4]} 6... Nd4 {[%clk 0:01:01]} 7. Nxd4 {[%clk 0:00:55.2]} 7... Bxd4 {[%clk 0:01:01.1]} 8. Nb5 {[%clk 0:00:54.4]} 8... Bc5 {[%clk 0:00:59.8]} 9. Bg2 {[%clk 0:00:52.9]} 9... a6 {[%clk 0:00:58.9]} 10. Nc3 {[%clk 0:00:48.8]} 10... c6 {[%clk 0:00:55.9]} 11. O-O {[%clk 0:00:46.4]} 11... d6 {[%clk 0:00:44.9]} 12. a4 {[%clk 0:00:42.3]} 12... Be6 {[%clk 0:00:44.7]} 13. Be3 {[%clk 0:00:20.6]} 13... Bxe3 {[%clk 0:00:40.2]} 14. fxe3 {[%clk 0:00:21.3]} 14... d5 {[%clk 0:00:39.5]} 15. d4 {[%clk 0:00:21.3]} 15... dxc4 {[%clk 0:00:35.2]} 16. dxe5 {[%clk 0:00:14.3]} 16... Nd7 {[%clk 0:00:32.8]} 17. h3 {[%clk 0:00:07.1]} 17... b5 {[%clk 0:00:25.9]} 18. a5 {[%clk 0:00:06.6]} 18... Nc5 {[%clk 0:00:23]} 19. b4 {[%clk 0:00:05.5]} 19... Nd3 {[%clk 0:00:19.4]} 20. Qg4 {[%clk 0:00:02.3]} 20... Bxg4 {[%clk 0:00:18.4]} 21. hxg4 {[%clk 0:00:03]} 21... Nxb4 {[%clk 0:00:18]} 22. e6 {[%clk 0:00:02]} 22... fxe6 {[%clk 0:00:17.4]} 0-1

How to represent movement?

Protocols	Type	Unambiguity	Human friendly
UCI(universal chess interface)	Only movement(DAG)	None	No(Engine)
SAN(standard algebraic notation)	Including Actions	Probably(Additional notation)	Yes(PGN)

How to tokenize it?



VS

```
> python data.py
-1  0  0  0  0  0  0  0
 0  0  1  0  0  0  0  0
 0  0  0  0  0  0  0  0
 0  0  0  0  0  0  0  0
 0  0  0  0  0  0  0  0
 0  0  0  0  0  0  0  0
 0  0  0  0  0  0  0  0
```

action10, knight!

https://github.com/yoonhero/sig_archive/tree/main/week06